Exercise 2

Consider split conformal prediction with data set $(X_1, Y_1), \cdots, (X_{n+2}, Y_{n+2}) \in \mathcal{X} \times \mathcal{Y}$ with given score $s : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$. We write $S_i = s(X_i, Y_i)$ for all i for simplicity. For $j \in \{1, 2\}$, the conformal p-value is the random variable defined by

$$p_j = \frac{1 + \sum_{i=1}^{n} \chi_{\{S_{n+j} \leqslant S_i\}}}{n+1}.$$

Given $\alpha \in (0, 1)$, let $\mathcal{C}$ be the conformal interval calibrated only on the first n observations. In other words, for $j \in \{1, 2\}$, the event $\{Y_{n+j} \in \mathcal{C}(X_{n+j})\}$ equals to $\{p_j < \alpha\}$. Assuming that $((X_1, Y_1), \cdots, (X_{n+2}, Y_{n+2}))$ is exchangeable, s is a symmetric score function, and that the random vector (p_1, p_2) satisfies the positive regression dependence on subsets (PRDS) assumption given in definition 10.3 of the textbook, prove that

$$P\{Y_{n+1} \in \mathcal{C}(X_{n+1}),\, Y_{n+2} \in \mathcal{C}(X_{n+2})\} \geqslant (1-\alpha)^2$$

without explicitly computing the probability on the left hand side of the inequality.

Solution First of all, Theorem 3.2 in the textbook already guarantees that, for $j \in \{1, 2\}$:

$$P\{Y_{n+j} \in \mathcal{C}(X_{n+j})\} \geqslant 1 - \alpha.$$

We focus on showing that

$$P\left\{Y_{n+2} \in \mathcal{C}(X_{n+2}) \mid Y_{n+1} \in \mathcal{C}(X_{n+1})\right\} \geqslant 1 - \alpha.$$

In order to make use of the PRDS condition, define a function g on the support of the discrete random variable p_1 by setting, for all t, that

$$g(t) = P\left(\{p_2 > \alpha\} \mid p_1 = t\right).$$

Note the following facts:

1. g is monotonically increasing because $g(t) = P\left(\{(p_1, p_2) \in A\} \mid p_1 = t\right)$ with the set A defined by all $(u, v) \in \mathbb{R}^2$ with $v < \alpha$, who is an increasing set.

2. The average value of $g(p_1)$ on the set $\{p_1 > \alpha\}$ is the desired conditional probability. In symbols:

$$\frac{1}{P\{p_1 > \alpha\}} \mathbb{E}\left[g(p_1); \{p_1 > \alpha\}\right] = P\left\{Y_{n+2} \in \mathcal{C}(X_{n+2}) \mid Y_{n+1} \in \mathcal{C}(X_{n+1})\right\}.$$

3. The expectation of $g(p_1)$ is the marginal covergae of (X_{n+2}, Y_{n+2}). In symbols:

$$\mathbb{E}\left[g(p_1)\right] = P\left\{Y_{n+2} \in \mathcal{C}(X_{n+2})\right\}.$$

4. The average value of $g(p_1)$ on the set $\{p_1 > \alpha\}$ is greater or equal to the average value of $g(p_1)$ on the set $\{p_1 \leqslant \alpha\}$ because g is monotonically increasing. That is:

$$\frac{1}{P\{p_1 > \alpha\}} \mathbb{E}\left[g(p_1); \{p_1 > \alpha\}\right] \geqslant \frac{1}{P\{p_1 \leqslant \alpha\}} \mathbb{E}\left[g(p_1); \{p_1 \leqslant \alpha\}\right]$$

The proof is complete once we combine the last three facts with the marginal coverage on (X_{n+2}, Y_{n+2}):

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\begin{aligned}
&1-\alpha \geqslant P\left\{Y_{n+2} \in \mathcal{C}(X_{n+2})\right\} = \mathbb{E}\left[g(p_1)\right] \\
&= \frac{P\{p_1 > \alpha\}}{P\{p_1 > \alpha\}} \mathbb{E}\left[g(p_1); \{p_1 > \alpha\}\right] + \frac{P\{p_1 \leqslant \alpha\}}{P\{p_1 \leqslant \alpha\}} \mathbb{E}\left[g(p_1); \{p_1 \leqslant \alpha\}\right] \\
&\geqslant (P\{p_1 > \alpha\} + P\{p_1 \leqslant \alpha\}) \frac{1}{P\{p_1 > \alpha\}} \mathbb{E}\left[g(p_1); \{p_1 > \alpha\}\right] \\
&= \frac{1}{P\{p_1 > \alpha\}} \mathbb{E}\left[g(p_1); \{p_1 > \alpha\}\right] = P\left\{Y_{n+2} \in \mathcal{C}(X_{n+2}) \mid Y_{n+1} \in \mathcal{C}(X_{n+1})\right\}.
\end{aligned}
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